

Phonon Engineering in Solid Polymer Electrolyte toward High Safety for Solid-State Lithium Batteries

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Extensively-used rechargeable lithium-ion batteries (LIBs) face challenges in achieving high safety and long cycle life. To address such challenges, ultrathin solid polymer electrolyte (SPE) is fabricated with reduced phonon scattering by depositing the composites of ionic-liquid (1-ethyl-3-methylimidazolium dicyanide, EMIM:DCA), polyurethane (PU) and lithium salt on the polyethylene separator. The robust and flexible separator matrix not only reduces the electrolyte thickness and improves the mobility of Li^+ , but more importantly provides a relatively regular thermal diffusion channel for SPE and reduces the external phonon scattering. Moreover, the introduction of EMIM:DCA successfully breaks the random intermolecular attraction of the PU polymer chain and significantly decreases phonon scattering to enhance the internal thermal conductivity of the polymer. Thus, the thermal conductivity of the as-obtained SPE increases by approximately six times, and the thermal runaway (TR) of the battery is effectively inhibited. This work demonstrates that optimizing thermal safety of the battery by phonon engineering sheds a new light on the design principle for high-safety Li-ion batteries.

1. Introduction

Lithium-ion batteries (LIBs) have important applications in portable electronic devices, energy storage systems, and electric vehicles because of high energy density and long service life.^[1,2] However, the traditional liquid electrolyte (LE) in LIBs may inevitably leak, burn, and even explode under extreme conditions, which seriously limits their wide applications. Therefore, improving the safety of LIBs has been paid more and more attention.^[3–5] Compared to LEs, solid-state electrolytes^[6–8] (SSEs) show lower cost, better processability, and much higher safety. The SSEs are generally categorized into inorganic solid electrolytes (ISEs)^[9–11] and solid polymer electrolytes (SPEs).^[12–14] The latter SPEs incorporated with polymer matrix and Li salts exhibit lightweight, flexibility, manufacturing scalability, and good adhesion to electrodes, which makes them more useful in practical applications. Nevertheless, SPEs may also decompose due to the side

reactions to trigger uneven heat dissipation and even burn during the charge and discharge process. As a result, the thermal safety of polymer solid-state batteries remains a huge challenge for wide application.^[15,16]

The thermal runaway (TR) of the battery is mainly rooted in the local heating induced by both the exothermic electrochemical reaction and the Joule heat formed from the charging-discharging process.^[17,18] However, at present, there is few research to optimize the thermal safety of batteries from the perspective of phonon scattering.^[19] It is well known that the thermal stability of polymer solid-state batteries is determined by thermal decomposition temperature of the polymer. However, in SPEs, the polymers are amorphous and often featured with low thermal conductivity (κ) (as shown in Figure 1a).^[20,21] Various attempts have been done to enhance the thermal properties of polymers by incorporating inorganic materials with high thermal conductivity, such as carbonaceous materials^[22] boron nitride,^[23] and metallic oxides^[24] to create additional heat pathways. However, due to the polymers' intrinsic properties like weak chain bonds, chain twisting, entanglement, chain ends, and extrinsic multiple scatterings, it is difficult to essentially improve the κ of polymer electrolytes.^[25] Thus, increasing the κ , which involves both electronic (κ_E) and lattice (κ_L) contributions,^[26] of the electrolytes

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DOI: 10.1002/adma.202405097

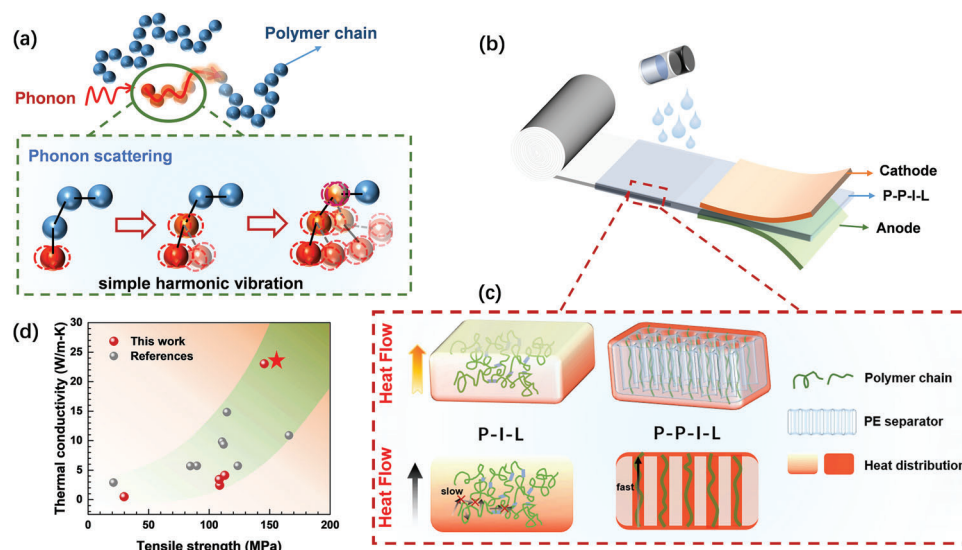


Figure 1. a) Thermal conduction mechanism in polymers. b) Schematic illustration of the facile process that can be adapted to assemble SSLBs with PPIL electrolyte; c) illustration of thermal transport paths in PIL/PPIL electrolytes. d) Comparison of the thermal conductivity and tensile strength of PIL/PPIL films with other polymers.^[21,25,47–49]

is an effective strategy to improve the internal thermal transport performance of the batteries. Furthermore, since most polymers are insulators with phonons as the dominant heat carriers, changing the chain bond is straightforward to enhance the thermal conductivity and to optimize the κ in polymers^{20,25}. According to the Boltzmann transport theory of phonons, κ_L is determined by the specific heat (C_V), phonon group velocity (v_g) and phonon relaxation time (τ), which are all strongly linked to the dispersion of phonons, including acoustic phonons and optical phonons. Compared to optical phonons,^[27,28] acoustic phonons are the main contributors to κ_L for having much higher velocities, and thus reducing the scattering rate of acoustic phonons is the key point for intensifying the κ_L and eventually improving the κ .^[29–31]

In addition to enhancing the thermal properties of SSEs, improving their thermal stability to solve the combustion risk is also a favorable solution to the TR of batteries.^[32,33] Introduction of flame retardants^[34,35] can effectively inhibit the combustion of SPEs, but the mechanical properties and electrochemical properties of polymer electrolytes might be deteriorated due to the different composition of flame retardants.^[36] Therefore, it is necessary to choose a flame retardant that can not only improve the safety of the SPEs, but also enhance the mechanical and electrochemical properties of the batteries. Ionic liquids (ILs)^[37,38] with intrinsic nonflammability and high conductivity, as well as the strong hydrogen bond and electrostatic interaction,^[39,40] are widely used in the optimization of SPEs. A nonflammable IL composed of 1-ethyl-3-methylimidazolium dicyanide (EMIM:DCA, has a high ionic conductivity of 2.8 S m⁻¹ at 25 °C) was used in this work, which the electrostatic interaction between the polymers and imidazole cations keeps IL stably locked in the dual network, allowing it to move freely with voltage.^[41,42] This not only improves the ionic conductivity of the SPEs, but also enhances its thermal stability by nitrogen-rich IL.

In this work, we focus on the influence of phonon scattering on SPEs, and design the SPE with high thermal stability by phonon engineering strategy to optimize the thermal safety of high-energy-density SSLBs. We propose a facile, large-scalable, and cost-effective strategy to synthesize ultrathin (≈ 13 μ m) and flexible SPEs for high-safety and high-energy-density solid-state lithium batteries (SSLBs). As shown in Figure 1b, PU/EMIM:DCA/LiTFSI (PIL) used as a model polymer electrolyte is infiltrated onto the PE separator matrix to prepare the hybrid SSE(PE/PU/EMIM:DCA/LiTFSI, PPIL). Herein, the introduction of EMIM:DCA breaks the random intermolecular attraction of the PU polymer chain and significantly reduces phonon scattering to promote the internal thermal conductivity of the polymer. Moreover, the robust and flexible separator matrix maintains the structural integrity of the ultrathin SSE,^[43–46] which not only prevents internal short circuit and dendrite penetration, but also importantly reduces external scattering and provides a relatively regular channel (Figure 1c). Thus, the thermal conduction of PPIL is greatly improved as compared with other SPEs^[21,25,47–49] (Figure 1d). In this work, we show a new way to enable LIBs with high safety and high energy density, and highlight the importance of the phonon engineering in high safety of LIBs.

2. Results and Discussion

We propose a facile, large-scalable, and cost-effective strategy to synthesize an ultrathin and flexible SSE. PU/EMIM:DCA/LiTFSI (PIL, with 40 wt.% LiTFSI), a polymer electrolyte used to transport Li⁺, was infiltrated into the PE separator matrix (with thickness of ≈ 9.1 μ m, Figure 2a) to prepare the hybrid SSE. The scanning electron microscope (SEM) images show that PIL evenly fills the nanopores of the PE separator (Figure S1, Supporting Information). The details on materials, calculations, characterizations, and electrochemical measurements are given in the

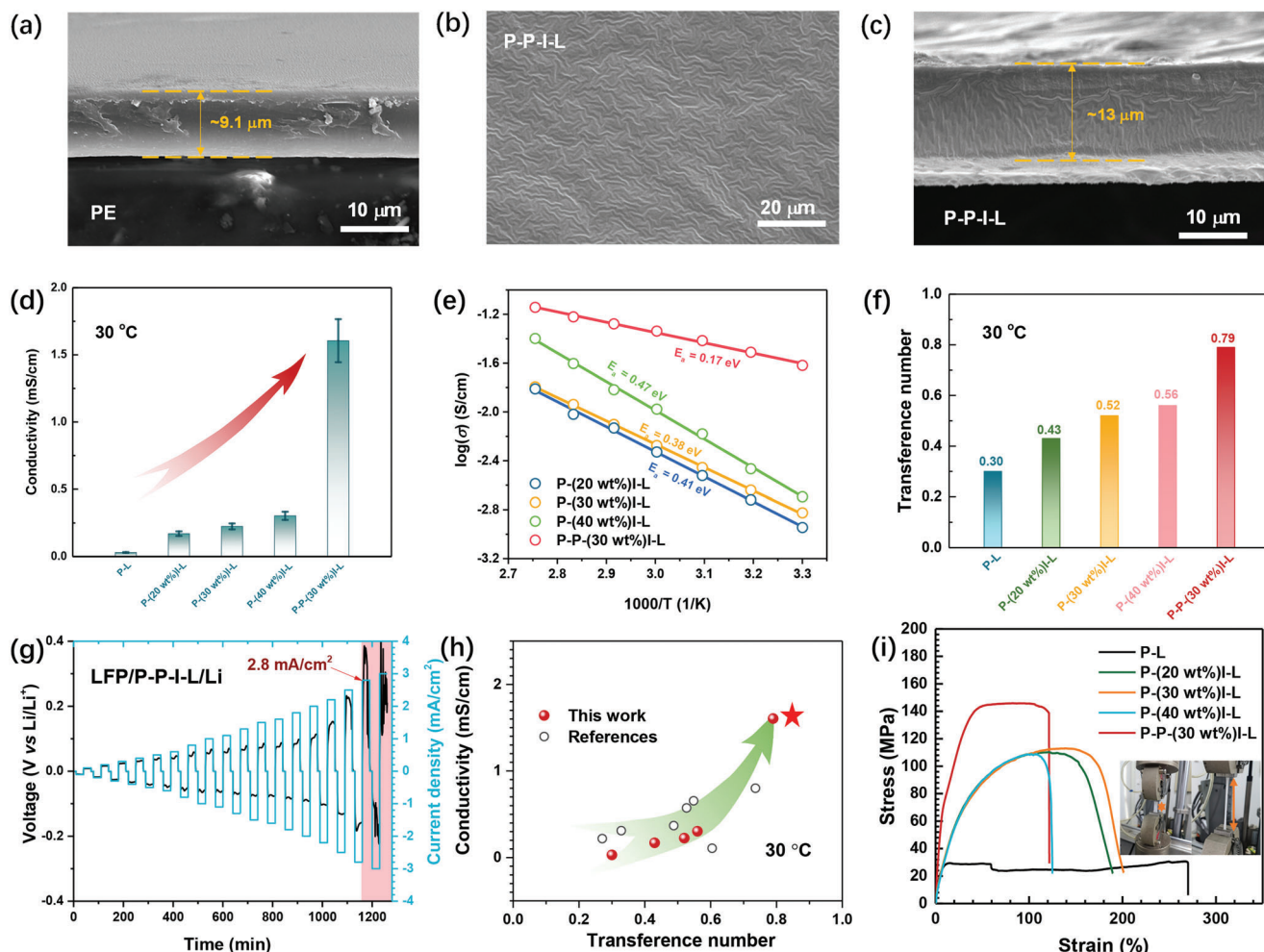


Figure 2. a) Cross-section SEM images of the PE separator. b) Top-section and c) cross-section SEM images of the separator after PIL infiltration. d) Conductivity of different components for PIL and PPIL electrolyte at room temperature. e) Arrhenius plots of the ionic conductivity data for PIL and PPIL electrolytes. f) Transference numbers of PL, PIL, and PPIL electrolytes. g) Critical current density of symmetric Li//Li cells paired with PIL and PPIL electrolytes. h) Comparison of transference number and ionic conductivity between this work and reported references.^[55–57] i) Stress–strain curves of PL, PIL, and PPIL electrolytes.

supporting information. In addition, the components of the following PPIL in this work are all PE/PU/(30 wt.%)EMIM:DCA/(40 wt.%)LiTFSI because of the best combined ionic conductivity and mechanical property with 30 wt.% EMIM:DCA. The powder X-ray diffraction (XRD) patterns for PILs (EMIM:DCA account at 20 wt.%, 30 wt.%, and 40 wt.%) are shown in Figure S2b (Supporting Information). From the diffraction peaks, we can see that the crystallinity of the polymer decreases with increasing EMIM:DCA content, and the amorphous phase of lithium salt is beneficial for ion conducting, which is further demonstrated by the secondary electron image of P(30 wt.%)IL (Figure 2b). To further confirm the PIL's crystal phase, the enlarged Fourier transform infrared (FTIR) spectra shows obvious peaks at 882, 814 and 689 cm^{-1} (Figure S2c, Supporting Information), suggesting a crystal form change of PILs to a polar phase with higher electric constant. High electric constant is favorable to the dissociation of lithium salt and the Li-ion conducting. Thus, compared with bare PU/LiTFSI elec-

trolyte, the addition of EMIM:DCA can effectively enhance the ion conducting ability and improve the electrochemical properties.

The thickness of SSE is very important, Li^+ diffusion time in SSE is proportional to the square of the electrolyte thickness, and the conductance of SSE is reversely proportional to its thickness.^[50,51] From the SEM images, it can be found that the film thickness slightly increases to $\approx 13 \mu\text{m}$ after PIL infiltration (Figure 2c), which is much thinner than other SSEs, implying excellent conductance and fast Li-ion migration.^[44] To verify this point, the ionic conductivity of SSEs was measured by electrochemical impedance spectroscopy (EIS) using stainless steel/SSE/stainless steel symmetric cells at a temperature range from 30 to 90 °C (Figures S3 and S4, Supporting Information). As shown in Figure 2d, the ionic conductivity of PPIL is 1.61 mS cm^{-1} at 30 °C, greatly higher than that of the pure PU/LiTFSI membrane (denoted as P-L, 0.03 mS cm^{-1} at 30 °C). Obviously, ultrathin PPIL delivers huge advantage on ionic conductance since PPIL ($\approx 13 \mu\text{m}$) processes significantly lower

thickness than PIL ($\approx 100 \mu\text{m}$). Also, the temperature dependence of ionic conductivity can be well described by Vogel–Tammann–Fulcher (VTF) behavior (Figure 2e). The activation energies of PPIL and PIL electrolytes are calculated by fitting the conductivity data using the VTF empirical equation^[50,52]:

$$\sigma = \sigma_0 T^{-1/2} \exp\left(-\frac{E_a}{R(T - T_0)}\right) \quad (1)$$

where E_a is the activation energy, σ_0 is the pre-exponential factor, T_0 is determined by the glass transition temperature, and R is the ideal gas constant. The fitting value of E_a for PPIL electrolyte is 0.17 eV, which is noticeably lower than that for PIL electrolyte (0.38 eV). The lower activation energy indicates that the ions need lower energy to move, demonstrating that Li ions are more easily transported in PPIL electrolyte than in PIL electrolyte, which may be attributed to the confinement of PU chains in nanochannels of the separator matrix.^[51,53] Li-ion transference number (t_{Li^+}) is another crucial parameter for the ionic transportation, which can be determined by the Bruce–Vincent method.^[54] As shown in Figure 2f, the PPIL electrolyte has the highest transference number ($t_{\text{Li}^+} = 0.79$), indicative of the strongest anion movement among three electrolytes during the lithium plating/stripping cycling process (more detailed information is shown in Figure S5, Supporting Information). In this work, PPIL shows excellent Li^+ transport performance as compared with recent reports (Figure 2h).^[55–57] Furthermore, because the ionic conductivity of PPIL electrolyte is higher than that of PIL electrolyte, the PPIL electrolyte displays a much higher critical current density of 2.8 mA cm^{-2} than that of 0.3 mA cm^{-2} for PIL electrolyte in the symmetric Li//Li symmetric-cells at room temperature (Figure 2g; Figures S6 and S7, Supporting Information). Interestingly, all the cells present overpotential protection instead of short circuit as the current density increases, suggesting the good mechanical strength and flexibility of the PPIL membrane in suppressing the penetration of Li dendrites. This can be further verified by the tensile test. As shown in the Figure 2i, PPIL presents a tensile module of 140 MPa and a large tensile strain of over 130%, indicating much excellent elastic module of PPIL polymer matrix, this is because of the hydrogen bonding and electrostatic interaction between the EMIM:DCA and PU matrix. In addition, the PE separator also makes an important contribution to improving the mechanical tensile properties of PPIL.

As expected, the symmetric cell assembled with PPIL electrolyte is stable at 0.2 mA cm^{-2} over 2000 h with low polarization ($\approx 80 \text{ mV}$, Figure 3a). Moreover, this symmetric cell is also stable at a fixed current density of 2 mA cm^{-2} over 600 h, which illustrates the stability of electrolyte against lithium metal (Figure S8, Supporting Information). SEM images also show the superior stability of PPIL electrolyte against lithium-metal anode (Figure 3b,c). After cycling at 0.2 mA cm^{-2} over 300 cycles, there is no obvious lithium dendrite appearing on the surface of cycled lithium metal in the cell with PPIL. This compact surface can reduce the inner corrosion of lithium metal during the plating/stripping process, which is consistent with the cross-section view of SEM images (Figure 3c). Ex situ depth XPS technique was carried out to investigate the reasons for the interfacial stability of PPIL electrolyte against lithium-metal anode. As shown in the Figure 3d, the peak intensity of F-Li species goes up after

200 cycles, suggesting that the presence of LiF can effectively passivate the reactive surfaces to reduce the continued electrolyte decomposition. LiF-rich SEI can also contribute to a fast Li^+ transportation kinetics and a blocked electron migration, and thus enhance the interfacial stability of PPIL against lithium metal. In addition, the low intensity of C-F species in the C1s spectrum indicates the presence of residual PIL and TFSI[−] debris on the lithium-metal anode. This debris can be attributed to the good adhesion situation between electrolyte and lithium metal, which benefits to reduce interfacial resistance. Furthermore, the differential charge density between TFSI[−] anions and PU/EMIM:DCA clearly proves the electron transfer from π -conjugation into the electrolyte environment (Figure 3e), in consistent with the electrostatic potential discussion above. In addition, density functional theory shows that the PU/EMIM:DCA presents higher binding energy (-1.464 eV) with TFSI[−] than that of -0.875 eV obtained from PU chains (Figure 3f), corresponding to the stronger adsorption of PU/EMIM:DCA on TFSI[−], which contributes to the enhancement of t_{Li^+} .

Due to the amorphous structure and vibration of the macromolecular chains in polymers, most polymers have extremely low thermal conductivity. Therefore, during the processes of charge/discharge, it was easy to cause the thermal inhomogeneity in the SPEs, resulting in heat accumulation, which would not only cause the reaction of electrolyte itself and with the electrodes to intensify, but also result in the uneven deposition of Li^+ . Thus, the excellent SPEs also need superior thermal transportation and thermal stability to ensure safe battery operation. Due to the amorphous structure and vibration of the macromolecular chains in polymers, most polymers have extremely low thermal conductivity ($\kappa = 0.1\text{--}0.5 \text{ W mK}^{-1}$).^[20,21] To improve thermal transportation, designing and altering the structure of the molecule and chain can optimize the orientation of the polymer chain to enhance the thermal conductivity. Furthermore, the thermal conductivity of the polymer is closely related to phonon scattering.^[20] The thermal conductivity of the polymer is calculated according to the Debye equation^[58]:

$$\kappa = \frac{1}{3} C_v v \bar{l} \quad (2)$$

where C_v is specific heat capacity per unit volume, v and $\bar{l}$ denote the phonon velocity and mean free path of phonons, respectively. Because the internal molecular arrangement of amorphous materials is short-range ordered and long-range disordered, the sub-mean free path is limited to a few cell spacing, so it is challenging to significantly improve the thermal conductivity of polymer.

In this work, the addition of a porous separator (Figure 1b) not only reduces the external scattering but more importantly provides a relatively regular channel for the thermal conduction of the polymers. Without affecting the crystallinity and chain length of the polymer, the phonon scattering is effectively reduced, and the thermal conductivity of the electrolyte is greatly improved as compared with that of the pure polymer electrolyte (Figure 1d). The thermal transport properties of PILs/PPIL at room temperature are shown in Figure 4a,b. It can be seen that the thermal conductivity of the polymers increases due to the introduction of EMIM:DCA (Figure S9, Supporting Information). Through the hydrogen bond interaction between the two

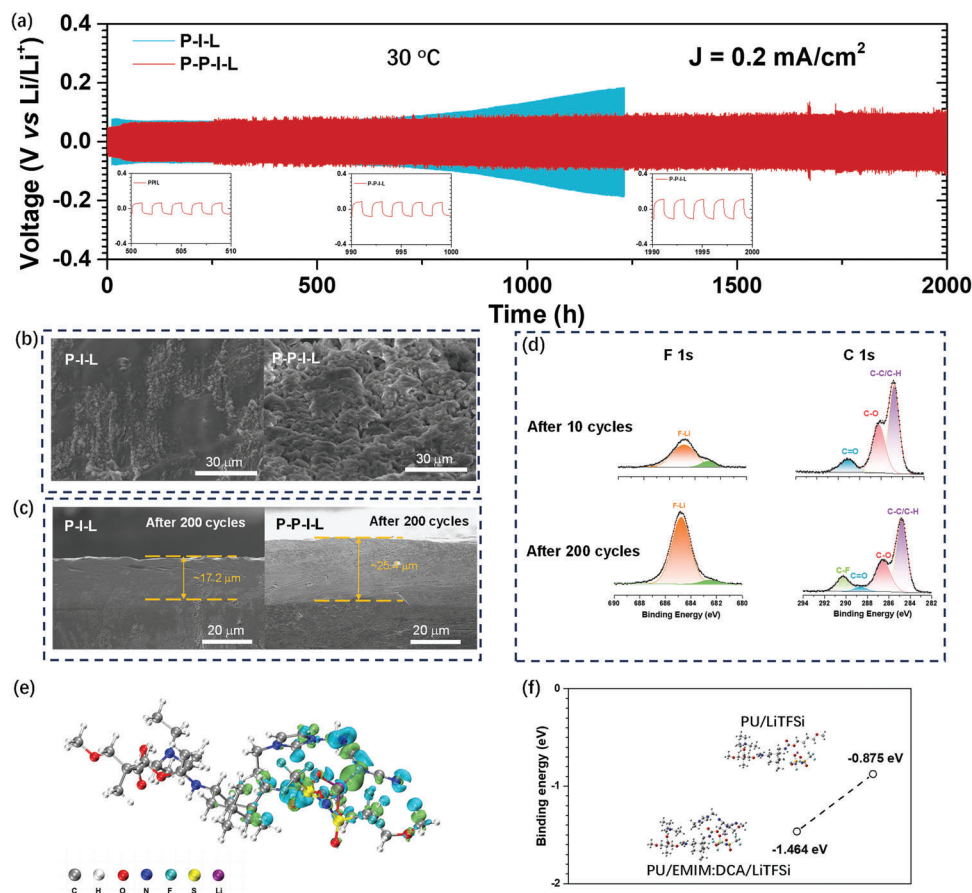


Figure 3. a) Voltage profiles of symmetric Li//Li cells with PIL or PPIL electrolyte at 0.2 mA cm^{-2} . b) Top view and c) cross-section view of SEM images of cycled lithium metal disassembled from Li/PPIL/Li and Li/PIL/Li cell. d) XPS spectra of cycled lithium anode disassembled from Li/PPIL/Li cell after cycled 200 s. e) Computed differential charge density distribution diagram of PU/EMIM:DCA on TFSI⁻, green color represents the accumulation of charge density, while the blue color indicates the reduction of charge density. f) The calculation of binding energy of TFSI⁻ with PU and PU/EMIM:DCA.

polymers, a new heat conduction pathway is created, thereby improving the thermal conductivity of the polymers. The addition of EMIM:DCA breaks the random intermolecular attraction of the PU polymer chain, and hence establishes an ordered and evenly distributed hydrogen bond and reduces phonon scattering to promote the internal thermal conductivity of the polymer. In addition, the excess introduction of EMIM:DCA might lead to its spontaneous polyreaction into polymerized ILs with chain entanglement structure, resulting in the enhancement of phonon scattering. That is the reason that we choose PE/PU/(30 wt.%)EMIM:DCA/LiTFSI, which shows the highest thermal conductivity at room temperature (Figure 4a), to prepare the PPIL electrolyte. Moreover, the thermal conductivity of PPIL is approximately six times larger than that of PIL (Figure 4b). This is because PU/EMIM:DCA/LiTFSI is confined to the nanopore of PE separator, which can also provide a neat thermal transport pathway to further help with heat dispersion. The specific heat of different electrolytes further demonstrates the reduction of phonon scattering in PPIL electrolyte (Figure S10, Supporting Information). More detailed thermal conductivity information of PILs is shown in Figure S8 (Supporting Information).

To determine the evolution of thermal transport properties of different PIL compositions, nuclear magnetic resonance (NMR)

spectra and Raman spectra were employed to analyze the molecular vibrations. The chemical structure was verified by ¹H chemical shifts of activated PILs (Figure S2d, Supporting Information), and the ¹H chemical shifts of PL and PIL gel electrolyte are basically the same, which indicates there is no change in the structure of PIL after introducing EMIM:DCA. From Raman spectra (Figure S11, Supporting Information), it is worth noting that all the peaks shift toward lower frequencies consistently, suggesting the increase in versus with the addition of EMIM:DCA. Meanwhile, broadening in the full width at half maximum for ν_5 provides important evidence of broadening in phonon dispersion for strong phonon scattering (Figure S11b, Supporting Information), confirming that the EMIM:DCA cationic disorder in PILs turns larger, which is in agreement with the structural analysis. These results further demonstrate that the thermal conductivity of SPEs should increase by adding flame retardant IL. In addition, thermogravimetric analysis (TGA) results display no weight loss at 350 °C (Figure 4c and Figure S12, Supporting Information), and PPIL has the smallest thermogravimetric ratio, which proved the excellent thermal stability. Furthermore, the differential scanning calorimetry (DSC) results illustrate the melting, thermal decomposition, and other processes of SPEs with temperature increasing. Herein, the reduction of peaks representing

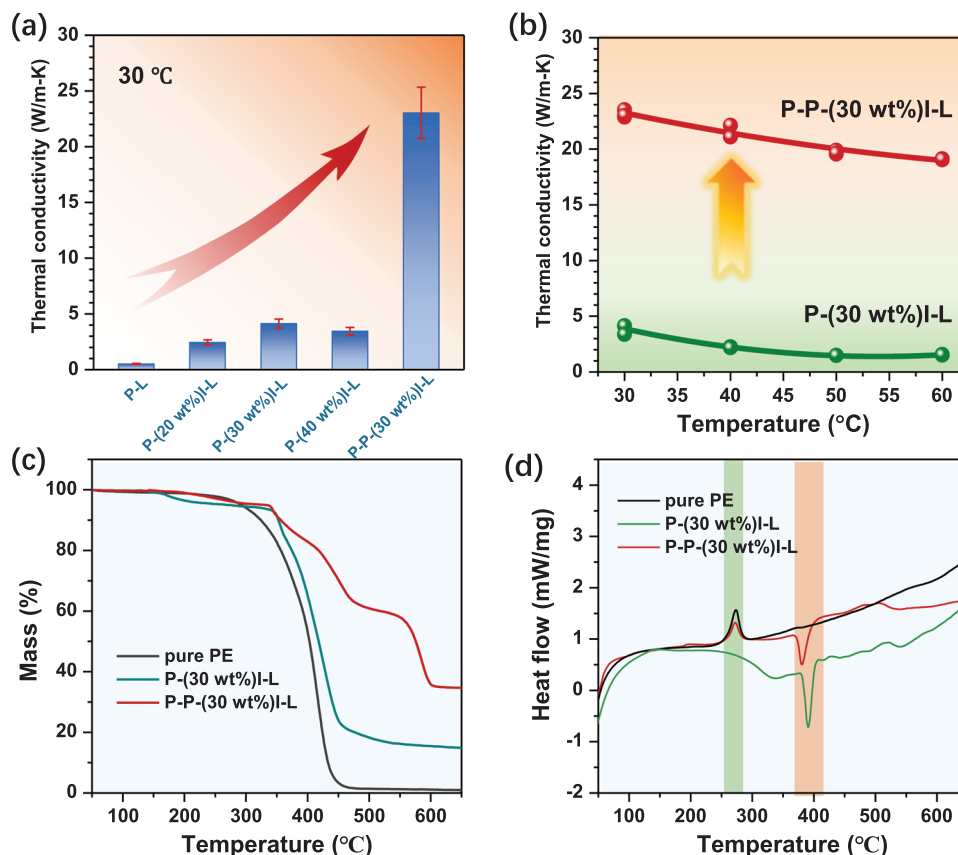


Figure 4. a) Thermal conductivities of PIL and PPIL at room temperature. b) Thermal conductivities of PIL and PPIL during 30–60 °C. c) TGA and d) DSC curves of PIL and PPIL during 50–650 °C.

melting of the PE separator (the green block) and the exothermic peaks representing the decomposition reaction of SPEs (orange block), also indicates excellent thermal stability of PPIL compared to PE and PIL.

To evaluate the electrochemical performance of ultrathin PPIL electrolyte under practical conditions, LFP/PPIL/Li full cells were assembled and tested at 30 °C. As shown in the **Figure 5a,b**, the cell assembled with PPIL electrolyte delivers discharge capacities of 168, 165, 160, 155, and 145 mAh g⁻¹ at 0.1, 0.2, 0.5, 1, and 2 C, respectively, and exhibits a high capacity retention after the current goes back to 0.2 C. By sharp contrast, Li/PIL/LFP cell presents inferior rate capability at high current densities, which discharge capacities are only 136 and 127 mAh g⁻¹ at 1C and 2C, respectively. Moreover, the cell with ultrathin PPIL electrolyte successfully delivers an initial discharge capacity of 165 mAh g⁻¹ and even maintains stable cycling at 30 °C and 0.2 C up to 300s (**Figure 5c,d**). This counterintuitive result is attributed to the lowered internal resistance endowed by the ultrathin nature of the PPIL electrolyte (**Figure 2d**; **Figures S3 and S4**, Supporting Information). The cell with ultrathin PPIL electrolyte delivers an initial discharge capacity of 157 mAh g⁻¹ and even maintains stable cycling at 30 °C and 0.5 C up to 200s (**Figure S13e,f**, Supporting Information). Compared to the battery with PIL, due to the shorter time it takes for Li⁺ to travel through the ultrathin SSE, the total resistance of the PPIL is evenly reduced by more than an order of magnitude. Furthermore, the voltage polarization has

no change in the Li/PPIL/LFP cell during cycling, indicating a stable impedance in the circuit of Li/PPIL/LFP cell, which coincides with the XPS results. Meanwhile, the high-voltage NCM811 cathode was also assembled with PPIL electrolyte to be the full cell Li/PPIL/NCM811, which further test the electrochemical performance of ultrathin PPIL electrolyte during high-voltage cycling process. As shown in **Figure S14** (Supporting Information), Li/PPIL/NCM811 delivers comparable rate capacities of 179, 168, 156, 133, and 81 mAh g⁻¹ at 0.2, 0.5, 1, 1.5, and 2 C with a charge cut-off voltage of 4.2 V, respectively. In addition, this cell can be still stably operated after current density goes back to 0.2 C. **Figure 5e** shows the charge-discharge profiles of Li/PPIL/NCM811 cell, which exhibits both high coulombic efficiency and excellent capacity reversibility. Li/PPIL/NCM811 cell delivers a comparable discharge capacity of 178 mAh g⁻¹ with a capacity retention of over 80% at 0.2 C after 300 cycles. This illustrates that, even under high-voltage, the ultrathin PPIL electrolyte can still normally operate because PPIL electrolyte has a wide electrochemical window of up to 4.8 V (**Figure S15**, Supporting Information).

To understand the effect of phonon scattering on TR mechanism, a series of ≈2.8 Ah pouch cells were evaluated by using accelerating rate calorimetry (ARC), as shown in **Figure 6a,b**. T_1 is defined as the onset temperature of self-heating, T_2 is the TR trigger temperature, and T_3 is the maximum temperature during TR, which are key parameters in the evaluation of the destructiveness

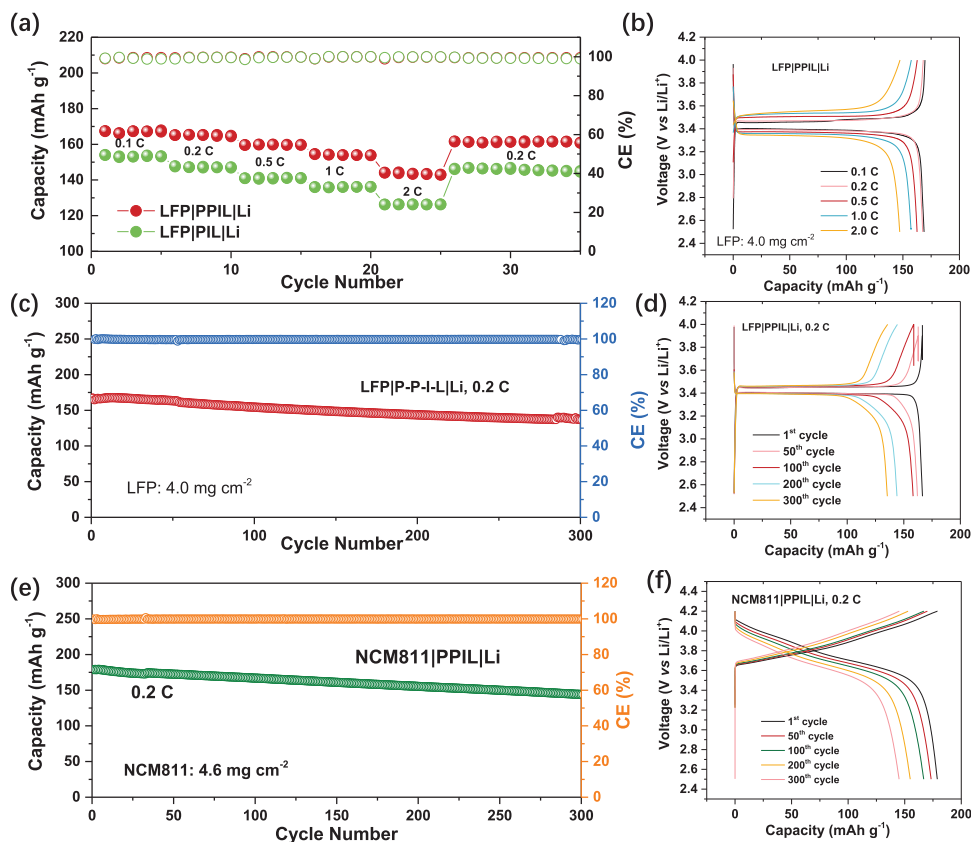


Figure 5. a) Rate performance and b) related voltage profiles of Li/PIL/LFP and Li/PPIL/LFP cell. c) Long cyclic performance and d) related voltage profiles of Li/PPIL/LFP cell at 0.2 C. e) Long cyclic performance and f) related voltage profiles of Li/PPIL/NCM811 cell at 0.2 C.

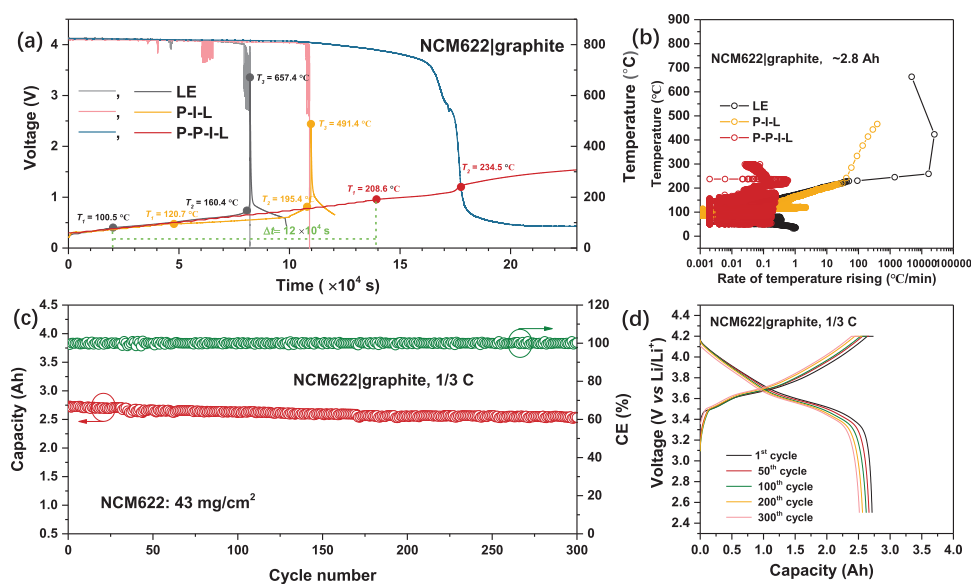


Figure 6. Intrinsic safety features of charged practical Ah-level pouch-type NCM622|graphite cells with LE, PIL, and PPIL electrolyte under ARC test: a) TR temperature plots and b) self-heating rate. c) Cycling stability and d) charge-discharge curves of typical NMC622 cell cycled during 2.5 V–4.2 V.

of TR^[59] It is critical to increase T_1 , maximize T_2 , and minimize T_3 for high-safety LIBs. The ARC behaviors are compared for the pouch cells with LE, PIL and PPIL electrolyte in Figure 6a,b. It can be seen that the inhibition effect of PIL electrolyte on TR is effective. Furthermore, by using PPIL, T_1 increases from 100.5 to 208.6 °C, and the occurrence of T_1 is delayed by $6 \times 10^{[4]}$ s, which indicates that PPIL electrolyte makes SEI more stable. Meanwhile, T_2 increases from 160.4 to 234.5 °C, and the delay in occurrence time of T_2 indicates that the PPIL electrolyte will decompose at a higher temperature, proving its high thermal stability. At the same time, T_3 disappears from 657.4 °C, implying that the PPIL electrolyte can effectively inhibit the TR due to the significantly reduced phonon scattering. Figure 6c,d shows the cycling performance of the cells. All cells exhibit initial discharge capacities up to 2.8 Ah at a rate of 1/3 C due to excellent electronic conduction, and their capacity retentions are $\approx 93\%$ even after 300 cycles. Therefore, a significant increase in thermal conductivity of the electrode materials induced by weak phonon scattering can effectively improve the thermal stability of the battery due to low heat generation and high-speed migration of ions.

3. Conclusion

A facile one-step solvent evaporation has been employed to fabricate an ultra-thin PE/PU/EMIM:DCA/LiTFSI polymer electrolyte with high conductivity and high thermal stability. The ultra-thin nature of the PPIL electrolyte greatly reduces cell resistance by shortening Li-ion diffusion distance. The robustness and flexibility provided by the separator matrix enhance the interfacial stability with electrodes and demonstrate high tolerance against abuses. The introduction of EMIM:DCA breaks the random intermolecular attraction of the PU polymer chain and thus reduces phonon scattering to promote the internal thermal conductivity of the polymer. More importantly, the addition of a porous separator not only reduces the external scattering but also gives a relatively regular channel for the thermal conduction of the polymers, and thus the thermal conductivity of PPIL is 6 times higher than that of PIL. As a result, both the coin lithium and pouch cells exhibit stable cycle performance at room temperature. It is proven that PPIL electrolyte can effectively inhibit the TR of the battery as compared to the LE. To summarize, our work demonstrates that phonon engineering can well optimize the thermal safety of rechargeable batteries, which sheds a new light on the design principle for high-safety SSLBs with high energy density.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (grant nos. 52027816, 52231009, 52102298). The authors thank the Analytical and Testing Center of Huazhong University of Science and Technology (HUST) and the State Key Laboratory of Materials Processing and Die & Mould Technology of HUST for characterization.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

phonon scattering, solid-state lithium batteries, solid polymer electrolyte, thermal runaway

Received: April 9, 2024
Revised: May 23, 2024
Published online: June 25, 2024

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